CHAPTER EXERCISE: Answers for this chapter start on page 314.

A calculator should NOT be used on the following questions.

1

In the xy-plane, what is the distance between the two x-intercepts of the parabola $y = x^2 - 3x - 10$?

A) 3
B) 5
C) 7
D) 10

2

What are the solutions to $x^2 + 4x + 2 = 0$?

A) $x = -2 \pm \sqrt{2}$
B) $x = 2 \pm 2\sqrt{2}$
C) $x = -2 \pm 2\sqrt{2}$
D) $x = -4 \pm 2\sqrt{2}$

3

If $a < 1$ and $2a^2 - 7a + 3 = 0$, what is the value of a ?

4

What is the sum of the solutions of $(2x - 3)^2 = 4x + 5$?

5

$$3x^2 + 10x = 8$$

If a and b are the two solutions to the equation above and $a > b$, what is the value of b^2 ?

A) $\frac{4}{9}$
B) $\frac{2}{3}$
C) 4
D) 16

6

$$f(x) = m\left[(x - m)^2 - 1\right]$$

In the function f defined above, m is a positive constant. The graph of f in the xy-plane is a parabola. Which of the following statements about the parabola is true?

A) Its minimum occurs at $(m, -1)$.
B) Its minimum occurs at $(m, -m)$.
C) Its maximum occurs at $(m, -1)$.
D) Its maximum occurs at $(m, -m)$.

7

$$y = -3$$
$$y = x^2 + cx$$

In the system of equations above, c is a constant. For which of the following values of c does the system of equations have exactly two real solutions?

A) -4
B) 1
C) 2
D) 3

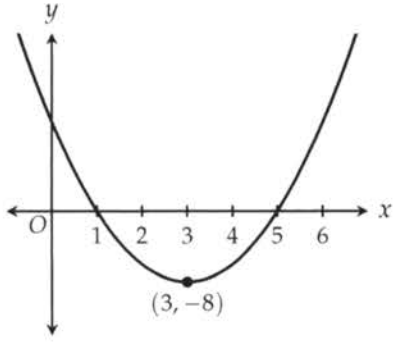
A calculator is allowed on the following questions.

8

At which of the following points does the line with equation $y = 4$ intersect the parabola $y = (x+2)^2 - 5$ in the xy-plane?

A) $(-1, 4)$ and $(-5, 4)$
B) $(1, 4)$ and $(-5, 4)$
C) $(1, 4)$ and $(5, 4)$
D) $(-11, 4)$ and $(7, 4)$

9



Which of the following equations represents the parabola shown in the xy-plane above?

A) $y = (x-3)^2 - 8$
B) $y = (x+3)^2 + 8$
C) $y = 2(x-3)^2 - 8$
D) $y = 2(x+3)^2 - 8$

10

For what value of t does the equation $v = 5t - t^2$ result in the maximum value of v ?

11

$$P = m^2 - 100m - 120,000$$

The monthly profit of a mattress company can be modeled by the equation above, where P is the profit, in dollars, and m is the number of mattresses sold. What is the minimum number of mattresses the company must sell in a given month so that it does not lose money during that month?

12

$$E(x) = 50x^2 - 800x + 10,000$$

The function above models the relationship between the total monthly expenses E, in dollars, of a restaurant and the number of tables x in its dining area, where $0 \leq x \leq 25$. What does the number 10,000 represent in the function?

A) The maximum number of tables that can fit in the dining area
B) The average monthly expenses, in dollars, for each table in the dining room
C) The total monthly expenses, in dollars, when there are zero tables in the dining area
D) The total monthly expenses, in dollars, when the number of tables in the dining area is maximized

13

$$f(x) = -x^2 + 6x + 20$$

The function f is defined above. Which of the following equivalent forms of $f(x)$ displays the maximum value of f as a constant or coefficient?

A) $f(x) = -(x-3)^2 + 11$
B) $f(x) = -(x-3)^2 + 29$
C) $f(x) = -(x+3)^2 + 11$
D) $f(x) = -(x+3)^2 + 29$

14

$$y = -3$$

$$y = ax^2 + 4x - 4$$

In the system of equations above, a is a constant. For which of the following values of a does the system of equations have exactly one real solution?

A) -4
B) -2
C) 2
D) 4

15

$$f(x) = x^2 - 24x + 180$$

For a manufacturer of x-ray machines, the cost per unit, in thousands of dollars, can be modeled by the function f above, where x is the weekly number of units produced. How many units should the manufacturer produce each week to minimize the cost per unit?

16

$$f(x) = -4x^2 + 22x$$

The function f above gives the data transfer speed, in megabytes per second, over a network connection x minutes after a file transfer was initiated. The graph of $y = f(x)$ in the xy-plane has x-intercepts at $x = 0$ and $x = b$. Which of the following is the best interpretation of b ?

A) The initial data transfer speed over the network connection
B) The maximum data transfer speed over the network connection
C) The time at which the data transfer speed over the network connection was at its highest
D) The time at which the file transfer completed

17

$$g(x) = -3x^2 + 18x$$

The function g above gives the data transfer speed, in megabytes per second, over a network connection x minutes after a file transfer was initiated. The graph of $y = g(x)$ in the xy-plane has x-intercepts at $x = 0$ and $x = c$. Which of the following is the best interpretation of the value of $\frac{c}{2}$?

A) The initial data transfer speed over the network connection
B) The maximum data transfer speed over the network connection
C) The time at which the data transfer speed over the network connection was at its highest
D) The time at which the file transfer completed

18

$$y = a(x - 3)(x - k)$$

In the quadratic equation above, a and k are constants. If the graph of the equation in the xy-plane is a parabola with vertex $(5, -32)$, what is the value of a ?

A) 2
B) 5
C) 6
D) 8

19

In the xy-plane, the line $y = 2x + b$ intersects the parabola $y = x^2 + bx + 5$ at the point $(3, k)$. If b is a constant, what is the value of k ?

A) 0
B) 1
C) 2
D) 3

Chapter 17: Quadratics

CHAPTER EXERCISE:

1. [C] We factor to find the x-intercepts.

$$y = x^2 - 3x - 10 = (x-5)(x+2)$$

The x-intercepts are 5 and -2. The distance between them is $5-(-2) = 7$.

2. [A] Using the quadratic formula,

$$x = \frac{-4 \pm \sqrt{(4)^2 - 4(1)(2)}}{2(1)} = \frac{-4 \pm \sqrt{8}}{2}$$
$$= \frac{-4 \pm 2\sqrt{2}}{2}$$
$$= -2 \pm \sqrt{2}$$

3. [.5]

$$2a^2 - 7a + 3 = 0$$
$$(2a-1)(a-3) = 0$$

If you had trouble factoring this, remember that you can always use the quadratic formula. Since $a < 1$, $a = \frac{1}{2}$, or 0.5.

4. [4] Expanding everything,

$$(2x-3)^2 = 4x + 5$$
$$4x^2 - 12x + 9 = 4x + 5$$
$$4x^2 - 16x + 4 = 0$$

The sum of the solutions is $-\frac{b}{a} = -\frac{-16}{4} = 4$.

5. [D] Move the 8 to the left side to get $3x^2 + 10x - 8 = 0$. Now, we can either use the quadratic formula or factor. In this case, we'll go with factoring: $(x+4)(3x-2) = 0$

So, $x = -4$ or $x = \frac{2}{3}$. Since $a > b$, b must be -4 and $b^2 = (-4)^2 = 16$.

6. [B] The minimum or maximum of a parabola always occurs at its vertex. Since m is positive, the parabola opens upwards in a "U" shape, which means we're dealing with the parabola's minimum at its vertex. Since expanding f gives us a vertex form of $f(x) = m\left[(x-m)^2 - 1\right] = m(x-m)^2 - m$, the vertex is at $(m, -m)$. Therefore, the parabola's minimum occurs at $(m, -m)$.

7. [A] Substituting the first equation into the second,

$$-3 = x^2 + cx$$
$$0 = x^2 + cx + 3$$

The system of equations will have two solutions if the equation above has two solutions. For the equation above to have two solutions, the discriminant, $b^2 - 4ac$, must be positive.

$$c^2 - 4(1)(3) > 0$$
$$c^2 - 12 > 0$$
$$c^2 > 12$$

Testing each of the answer choices, only answer (A), -4, gives a value bigger than 12 when squared.

8. [B] To find the intersection points, treat the two equations as a system of equations. Substituting the first equation into the second,

$$4 = (x+2)^2 - 5$$
$$9 = (x+2)^2$$
$$\pm 3 = x + 2$$
$$x = -5, 1$$

The y-coordinates of the intersection points must be 4 (from the first equation), so the two points of intersection are $(-5, 4)$ and $(1, 4)$.

9. [C] Because the vertex is at $(3, -8)$, the answer must be either (A) or (C). Because the parabola passes through $(1, 0)$, we can use that point to test out our two potential answers. When we plug in $x = 1$ into (C), we get $y = 0$, confirming that the answer is (C).

10. [2.5] From the equation $v = 5t - t^2 = t(5 - t)$, we can see that the t-intercepts are 0 and 5. Because the maximum occurs at the vertex, whose t-coordinate is the average of the two t-intercepts, $t = 2.5$ results in the maximum value of v. You can confirm this by graphing the equation on your calculator.

11. [400] To find the minimum number of mattresses the company must sell so that it doesn't lose money, set $P = 0$.

$$\begin{aligned} m^2 - 100m - 120,000 &= 0 \\ (m - 400)(m + 300) &= 0 \\ m &= -300, 400 \end{aligned}$$

Since it doesn't make sense for the number of mattresses sold to be negative, $m = 400$. If you had trouble factoring the equation above (it's tough), graphing on your calculator and the quadratic formula are both good alternatives.

12. [C] The number 10,000 is the y-intercept, the total monthly expenses when x, the number of tables, is 0. We can assume these expenses to be rent, equipment, worker salaries, etc.

13. [B] We need to complete the square. First divide everything by -1,

$$-y = x^2 - 6x - 20$$

Now divide the middle term by 2 to get -3 and square that result to get 9. We put the -3 inside the parentheses with x and subtract the 9 at the end.

$$-y = (x - 3)^2 - 20 - 9$$

Now simplify and multiply everything back by -1.

$$y = -(x - 3)^2 + 29$$

14. [A] Substitute the first equation into the second,

$$\begin{aligned} -3 &= ax^2 + 4x - 4 \\ 0 &= ax^2 + 4x - 1 \end{aligned}$$

For the system to have one real solution, the equation above should have only one real solution. In other words, the discriminant, $b^2 - 4ac$, must equal 0.

$$\begin{aligned} (4)^2 - 4(a)(-1) &= 0 \\ 16 + 4a &= 0 \\ 4a &= -16 \\ a &= -4 \end{aligned}$$

15. [12] We're looking for the value of x that results in the minimum value of $f(x)$. Since the graph of f is a parabola that opens upwards in a "U" shape, the minimum of f occurs at the vertex, which is located at $x = -\frac{b}{2a} = -\frac{-24}{2(1)} = 12$. Therefore, the manufacturer should produce 12 units each week to minimize the cost per unit.

16. [D] Since $x = 0$ and $x = b$ are x-intercepts, $f(x)$, the data transfer speed, is 0 at $x = 0$ and $x = b$. First, why would the data transfer speed be 0 at $x = 0$? Well, the file transfer is just starting so no megabytes have been transferred yet. Now why would the speed be 0 at $x = b$? The best answer is that the file transfer has just completed, so there are no more megabytes of data left to transfer—just like a car's speed would be 0 when it stops at the end of a trip. Therefore, b most likely represents the time at which the file transfer completed.

17. [C] Since $g(x)$ is a parabola and $x = 0$ and $x = c$ are its x-intercepts, $x = \frac{c}{2}$ is the parabola's axis of symmetry, the line along which the vertex lies. In this case, the vertex is where the maximum occurs, since the parabola opens downwards in an upside-down "U" shape. Therefore, $\frac{c}{2}$ is the time at which the data transfer speed was at a maximum.

18. D One of the x-intercepts is 3. Since the x-coordinate of the vertex, 5, must lie at the midpoint of the two x-intercepts, the other x-intercept is 7. Therefore, $k = 7$, giving us $y = a(x-3)(x-7)$. We can now plug in the vertex as a point to solve for a.

$$\begin{aligned} -32 &= a(5-3)(5-7) \\ -32 &= a(2)(-2) \\ -32 &= -4a \\ a &= 8 \end{aligned}$$

19. C Substituting the point $(3, k)$ into both equations,

$$\begin{aligned} k &= 2(3) + b \\ k &= (3)^2 + 3b + 5 \end{aligned}$$

This is a system of equations. Substituting the first equation into the second,

$$\begin{aligned} 2(3) + b &= (3)^2 + 3b + 5 \\ 6 + b &= 9 + 3b + 5 \\ 6 + b &= 3b + 14 \\ -8 &= 2b \\ b &= -4 \end{aligned}$$

From the first equation,

$$k = 6 + b = 6 - 4 = 2$$